

Sakshi Mohta

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SKILLS

Laboratory & Assay Techniques: PCR, SDS-PAGE, ELISA

Data & Programming: Python, R, SQL, MS Excel, Jupyter Notebook, Google Colab

Biomedical Engineering & Analysis: Data Analysis, Biomechanics, Medical Imaging, Systems Biology, Drug Discovery

Modeling & Simulation Tools: Computer-Aided Design (CAD), ANSYS Workbench

Professional & Research Skills: Experimental Design, Scientific Writing, Teamwork

EDUCATION

Master of Science (MS), Bioengineering, UC San Diego Expected 2027
San Diego, CA
Activities & Societies: IEEE, Data Science Student Society (DS3), Society of Women Engineers (SWE), Bioengineering Graduate Society (BEGS), Association for Women in Maths (AWM), Gender Minorities in Bioengineering (GMBE)

Bachelor of Technology (B.Tech), Biotechnology, Delhi Technological University Jul 2024
Delhi, India
• GPA: 3.75/4.0 | Activities & Societies: DTU Girls Chess Society, DelTech MUN Debate Society
• Relevant Coursework: Biochemistry, Computational Biology, Drug Design & Delivery, Heat & Mass Transfer, Instrumentation in Biotechnology, Polymer Waste Management, Stem Cells & Regenerative Medicine

EXPERIENCE

Python Data Analysis Intern, TechMedBuddy Apr 2025 – May 2025
Remote
• Processed and visualized large-scale biological datasets (FASTA, CSV, NGS outputs) to support experimental learning using Python libraries such as Pandas and Matplotlib for data cleaning, transformation, and analysis
• Developed Python modules to automate BLAST sequence processing for biomedical datasets, improving research data throughput and reproducibility
• Integrated domain-specific bioinformatics tools (e.g., NCBI BLAST API) to support prototype analysis workflows

Bioinformatics Intern, HackBio Feb 2025 – Mar 2025
Remote
• Collaborated in a team of 5 to develop Python-based tools, including RNA codon-to-protein translation algorithms
• Built and simulated logistic growth models to analyze biological system behavior, generating 100+ growth curves and calculating time-to-capacity metrics for quantitative performance evaluation
• Performed statistical analysis and data visualization on microbial growth datasets (OD600), using Welch's t-test and comparative plots to identify significant differences between biological variants
• Integrated and analyzed protein mutation datasets (SIFT and FoldX) to identify high-impact structural and functional variants

Undergraduate Researcher, Molecular Neuroscience and Functional Genomics Laboratory, Delhi Technological University Nov 2023 – May 2024
Delhi, India
• Built and analyzed large-scale biological networks (160+ genes, 900+ interactions) using Cytoscape, STRING, DGIdb, and DrugBank, applying centrality metrics and pathway enrichment (KEGG, GO) to identify hub genes
• Prioritized 5 candidate drugs using Connectivity Map (CMap) transcriptomic similarity analysis, identifying high-confidence repurposing candidates and validating findings via clinical trial and literature evidence
• Applied data-driven statistical and systems biology methods to integrate multi-omics datasets, supporting evidence-based decision-making in CNS drug discovery

Research Trainee, Helix Biogenesis Pvt. Ltd. Jun 2023 – Aug 2023
Noida, India
• Conducted hands-on training in molecular biology techniques including PCR, gel electrophoresis, and ELISA, gaining expertise in experimental design and optimization
• Assisted in molecular diagnostics workflows, learning industry-standard techniques for DNA/RNA extraction, quantification, and sequencing methodologies

AWARDS AND ACHIEVEMENTS

- Awarded the Zolve Global Need-Based Scholarship, selected as 1 of 30 recipients across India for funding graduate studies in the U.S.
- Achieved All India Rank 804 in GATE, Biotechnology, among 17,500 candidates
- Recognized with the Dr. Ram Prakash Malik Memorial Excellence Award—highest marks in CBSE XII Biology (99/100)
- Ranked 24th out of 450,000 participants in the National Science Concoors, Group Unicorn

CERTIFICATIONS

Fundamentals of Data Analysis in Precision Medicine & Cloud Computing, Stanford Deep Data Research Center Jan 2026

VOLUNTEERING

Earth5R, Biotechnology Content Writer Jan 2025 – Mar 2025
• Contributed to a UNESCO-endorsed sustainability initiative by creating content on biotechnological advancements, focusing on environmental impact

Muskurahat Foundation, Crowd Funding Intern Jun 2024 – Jul 2024
• Raised INR 15,100 to support quality education of underprivileged children in India